

Problem Statement, Articulation of Value, Potential Economic Value

The problem statement that this project seeks to answer is whether a predictive model can accurately estimate an individual customer's likelihood of responding to a future marketing offer. This project addresses that problem by building a model on demographic information, historical spending behavior, and prior campaign engagement, allowing a business to allocate its marketing budget toward the customers most likely to respond.

The value that this project seeks to articulate is the return a business gains by targeting its marketing spend rather than distributing it evenly across its entire customer base. This is significant as it directly improves the return generated by every dollar spent on a campaign, since a larger share of that spend reaches customers who are actually likely to convert. Without this kind of targeting, businesses continue to contact customers who have little interest in an offer, which wastes budget and risks fatiguing customers who are contacted too often with messages that don't apply to them. The value this project provides, then, is a way to direct marketing resources toward the customers most likely to respond, while also protecting the customer relationship by reducing unnecessary outreach. This will help businesses get more value out of every marketing dollar and free up budget that can be reinvested into higher-value engagement with their best customers.

The economic value of this project, therefore, is the net financial benefit a company gains by using a predictive model to target its marketing campaigns, as opposed to contacting its entire customer base. This value will be calculated by comparing two scenarios: a baseline scenario, in which a company contacts its entire customer base uniformly, and a targeted scenario, in which the company contacts only the segment of customers the model identifies as most likely to respond.

For a sample analysis, assume a customer base of 2,000, a baseline response rate of 15 percent, a cost of \$2.00 to contact each customer, and \$150 in revenue generated by each accepted offer. Under the baseline scenario, contacting all customers costs \$4,000 and generates \$45,000 in revenue from 300 responders, for a net value of \$41,000 and a return of roughly \$10 for every dollar spent. Under the targeted scenario, the model contacts only the top 40 percent of customers by predicted likelihood, capturing approximately 75 percent of true responders within that group. This reduces cost to \$1,600 while still generating \$33,750 in revenue from 225 responders, for a net value of \$32,150 and a return of roughly \$20 for every dollar spent — about double the return of the baseline scenario, along with \$2,400 in freed-up budget the company can reinvest elsewhere.

Metric	Base Scenario	Targeted Scenario
Customers contacted	2,000	800 (top 40%)
Response rate	15%	75% of true responders
Responders reached	300	225
Total cost	\$4,000	\$1,600
Total revenue	\$45,000	\$33,750
Net economic value	\$41,000	\$32,150
Return per dollar spent	~\$10	~\$20
Budget freed for reinvestment	N/A	\$2,400

Dataset and Modeling Discussion

The dataset that this project will use to answer the problem statement is the [Customer Personality Analysis dataset](#), sourced from Kaggle. This dataset works because it contains exactly the categories of information the problem statement calls for: demographic information such as income, education, and household composition; historical spending behavior across product categories over a two-year period; and prior campaign engagement, recorded as whether each customer accepted each of several past marketing offers. It will solve the problem statement by training a model on these attributes alongside a confirmed outcome for each customer — whether that customer accepted the most recent marketing campaign — using its 2,240 labeled records to learn which combinations of income, spending, and prior engagement are actually associated with a customer responding. Once trained, the model can then be applied to new customers to estimate their likelihood of responding before any offer is actually sent.

The type of modeling that will solve this problem is supervised learning, since every customer record in the dataset includes a labeled outcome indicating whether that customer accepted the offer. The modeling problem this project addresses, then, is a supervised classification problem, and specifically a binary classification problem, since the outcome being

predicted takes only two possible values: whether a customer accepts the offer or not. This will allow the project to directly estimate each customer's probability of responding, which is the precise prediction the problem statement requires.

13 Week Project Plan

The project will follow an end-to-end machine-learning process, beginning with problem definition and concluding with a finalized model, report, GitHub repository, and presentation.

WEEK 1 - Problem Definition

Define the business problem as predicting which customers are likely to respond to a marketing campaign, so a company can target outreach efficiently instead of contacting its entire customer base. Establish the project's scope, objectives, business value, team responsibilities, and GitHub repository.

WEEK 2 - Data Collection and understanding

Download the Customer Personality Analysis dataset and import it into Python. Document the dataset's source, dimensions, variables, and limitations. Classify the variables into demographic, household, spending, purchasing-channel, and campaign-response categories.

WEEK 3 - Data-quality assessment

Examine data types, missing values, duplicate customer records, invalid categories, and unrealistic observations. Investigate potential outliers in variables such as income, birth year, and product spending.

WEEK 4 - Data Cleaning

Address missing values and determine how to handle unreasonable values and influential outliers. Convert the enrollment-date field into a usable date format and standardize inconsistent categories in education and marital status.

WEEK 5 - Exploratory data analysis

Analyze customer demographics, income, recency, product spending, purchasing channels, household composition, and campaign acceptance, with particular attention to how these

variables differ between customers who accepted the most recent campaign and those who did not. Create visualizations showing important distributions, relationships, and behavioral patterns.

WEEK 6 - Feature engineering

Create useful customer-level measures such as age, customer tenure, total spending, total purchases, total children, prior campaign acceptances, and preferred purchasing channel. Remove identifiers and variables that do not contribute meaningfully to predicting campaign response.

Week 7 - Data preparation and segmentation

Encode categorical variables and scale numerical features. Build an initial K-means clustering model on the prepared customer features to surface natural customer segments — this segmentation will support interpretation of the classifier's predictions later, rather than serving as the project's primary model.

WEEK 8 - Train/Test Split and Baseline Classification Model

Split the data into training and test sets using the campaign-response outcome as the target label. Train a baseline classification model (such as logistic regression) to predict whether a customer will respond to a future campaign, and establish baseline performance.

WEEK 9 - Alternative modeling approaches

Compare the baseline model against at least one alternative classification approach, such as a decision tree, random forest, or gradient boosting model. Address class imbalance in the response variable using techniques such as class weighting or resampling.

WEEK 10 - Model Evaluation and Selection

Compare candidate models using classification metrics such as precision, recall, F1-score, and ROC-AUC, rather than raw accuracy, given the imbalance in campaign response. Select the model that best balances correctly identifying likely responders against minimizing wasted outreach.

WEEK 11 - Customer-segment profiling

Examine which features drive the selected model's predictions, and connect these findings back to the customer segments identified in Week 7 — for example, identifying which segments are disproportionately represented among predicted responders. Assign clear descriptive context to these findings to support the business recommendations that follow.

WEEK 12 - Marketing recommendations and economic value

Develop a marketing strategy that uses the model's predictions to prioritize outreach, including suitable products, channels, and offers for high-likelihood customers. Estimate the potential economic value of model-informed targeting using documented assumptions about campaign costs, response rates, customer spending, and audience size.

WEEK 13 - Final documentation and presentation

Finalize the report, Python notebooks, visualizations, model evaluation results, segment interpretations, recommendations, and conclusions. Organize and test the GitHub repository to ensure the analysis is reproducible, document all assumptions and sources, and prepare the final project presentation and demonstration.